School Management System

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***Abstract*— The School Management System project aims to address manual and time-consuming administrative processes in educational institutions by developing a comprehensive software solution. This system will streamline tasks such as student and teacher enrollment, record-keeping, and communication. Functionalities include user authentication, dashboards for administrators, and adding/managing student and teacher records. Non-functional requirements ensure system reliability, scalability, and data integrity. By creating a user-friendly interface and implementing robust security measures, the project seeks to enhance efficiency, promote transparency, and improve collaboration among administrators.**

***Keywords*— flexibility, reporting, management software, Efficiency**

I. INTRODUCTION

Technologies are changing day by day. Technologies are playing main role in Airport2 bus station, Hospitals and etc. these days many Schools using School management system for good and efferent management. but in Sri Lanka few schools are using School management system. Most schools in this country have system of computers but they are only used for teaching the "Information Technology subject and nobody uses them for Management purposes. Some restricted numbers of schools are using Microsoft word, Microsoft Excel for their few Administrations and management proposes. How ever there are numerous tasks that can be done by computers. but they are manually calculating everything finally using computers for designing printing purpose only.

Schools may struggle with manual and time-consuming administrative processes such as student enrollment, teacher enrollment and record-keeping. This can lead to inefficiencies, errors, and a strain on staff resources. Schools may find it difficult to manage and analyze vast amounts of student data, Without proper systems. Also, schools risk compromising the privacy and confidentiality of personal data. And they faced Ineffective communication between teachers, students, and administrators.

Therefor I am implementing school management system for lese cost, overcome with some existing problems. My school management system will have two main parts Students, Teachers management system. It provides functionalities such as user authentication, dashboards for administrators, adding and managing students and teachers. Those are most difficult tasks for management. This school management system is designed for helping to management.

II. LITERATURE REVIEW

In this section, we review existing systems similar to our School Management System (SMS) to understand their features, strengths, and weaknesses. This comparison will inform the design and development of our system to ensure it meets the needs of educational institutions effectively. The project seeks to enhance the efficiency and precision of managing information regarding the Student and Teacher, replacing the manual method that often leads to errors and disruptions in school record. This literature review provides an overview of the existing research on automated school bell systems and their benefits.

A. Teachers Record Management System Project Report

Acharya (2023) explores a Teachers Record Management System aimed at effectively organizing and managing teacher-related data. The system focuses on functionalities such as personal information management, attendance tracking, and performance evaluations. It highlights the benefits of digital records for enhancing data accuracy, streamlining administrative tasks, and facilitating decision-making processes [1]. (Acharya, 2023)

B. Online Bus Reservation System

Acharya (2024b) presents an Online Bus Reservation System, focusing on simplifying the booking process for bus travel. The system offers features such as route management, seat selection, and payment processing. The paper highlights the benefits of online reservations, including convenience, time savings, and real-time updates on bus schedules and seat availability, alongside security measures for data protection [2]. (Acharya, 2024b)

C. *Online Shopping Management System*

Acharya (2024c) examines the development of an Online Shopping Management System aimed at enhancing user experience in e-commerce. The system's key functionalities include product catalog management, shopping cart integration, and order processing. The study addresses challenges like user authentication, data privacy, and system scalability, offering insights into optimizing backend processes to handle high transaction volumes [3]. (Acharya, 2024c)

D. *Effective School Management*

Everard, Morris, and Wilson (2004) provide a comprehensive guide on effective school management, covering aspects such as leadership, resource allocation, staff management, and policy implementation. They emphasize strategic planning and continuous improvement to achieve educational goals, discussing the role of technology in enhancing school management efficiency [4]. (Everard et al., 2004)

E. *Classification of the Features in Learning Management Systems*

Jurado et al. analyze the features of Learning Management Systems (LMS), classifying functionalities such as content delivery, communication tools, assessment mechanisms, and administrative capabilities. The study highlights how these features support instructional delivery, student engagement, and performance evaluation, underscoring the importance of selecting appropriate LMS features for specific educational needs [5]. (Jurado et al., n.d.)

Acharya (2023, 2024b, 2024c)[1]-[3]consistently emphasizes the role of centralized systems in improving data accuracy, streamlining administrative tasks, and enhancing overall efficiency. And Everard, Morris, and Wilson (2004) focus on strategic planning and continuous improvement to enhance school management efficiency.

Based on the insights from the literature, implementing a School Management System for less cost and overcoming existing problems involves addressing both student and teacher management. This includes functionalities such as user authentication, administrative dashboards, and management of student and teacher data. These tasks are critical for school management, aiming to enhance efficiency and support educational objectives. Ensuring the system is user-friendly and accessible, Balancing functionality with affordability, unifying various administrative tasks into a single system and protecting sensitive information and ensuring data integrity.

III. METHODOLOGY

The methodology for developing the School Management System (SMS) aims to address the challenges faced by educational institutions in managing administrative processes, student data, and communication effectively. By implementing an efficient and user-friendly system, schools can streamline operations, improve data accuracy, and enhance communication between stakeholders. This methodology outlines the key steps involved in the development and implementation of the SMS to ensure its success and effectiveness in meeting the needs of educational institutions.

Educational institutions face significant challenges in managing administrative tasks, academic records, and communication due to manual and fragmented processes. This inefficiency leads to data inaccuracies, delayed decision-making, and increased workload on staff. The problem is to design and develop an integrated School Management System that addresses these challenges by providing a comprehensive solution for managing student and teacher information, academic records. It is beginning with a thorough needs assessment, involving interviews and surveys with school stakeholders to understand existing challenges and requirement. The subsequent phases encompass system design and development, creating a comprehensive architecture for the scheduling software and automated hardware. User training and documentation facilitate a smooth transition, leading to system deployment. Ongoing support and maintenance ensure continued functionality, and user feedback informs iterative improvements. This approach ensures the systematic implementation of the School Management System, addressing specific school needs and delivering enhanced efficiency and precision in record management.

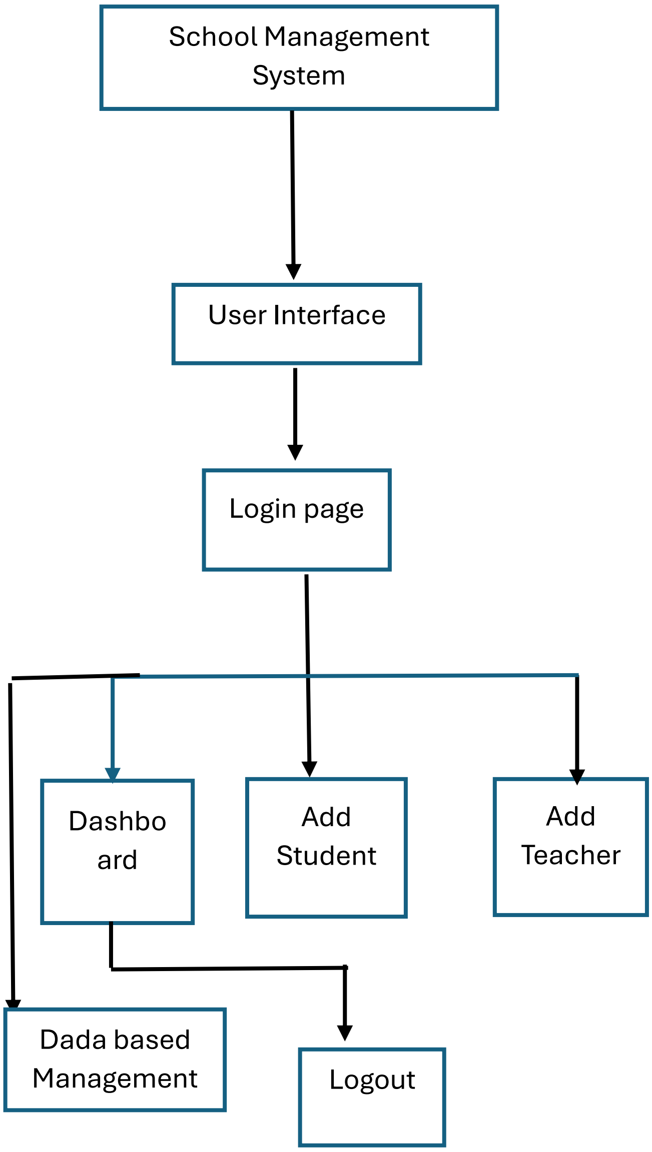

Fig.1 High-level architecture diagram of the methodology

A. Need Assessment

Conduct surveys and interviews with stakeholders (administrators, teachers and students) to identify pain points and requirements. Analyze existing processes and systems to understand the challenges and opportunities for improvement.

B. System Design and Development

Design the architecture and functionalities of the SMS based on the identified requirements to develop the core modules including user authentication, dashboards, student management, and teacher management Utilize technologies such as C# and SQL Server for development.

C. Installation and Integration

Set up the necessary hardware and software infrastructure for deploying the SMS. Integrate the SMS with existing system such as school databases or learning management systems.

D. Testing and Quality Assurance

Conduct thorough testing of the SMS to ensure functionality, performance, and security. Perform unit testing, integration testing, and user acceptance testing (UAT) to identify and fix any issues.

E. User Training and Documentation

Provide comprehensive training sessions for administrators, teachers, and other users on how to use the SMS effectively. Develop user manuals and documentation to guide users through the functionalities and features of the SMS.

F. Ongoing Support and Maintenance

Establish a support system to address any issues or concerns raised by users. Regularly update and maintain the SMS to ensure compatibility with evolving technologies and changing needs of educational institutions.

IV. DATA

A. Data Collection

Data was collected from a mid-sized school over a period of three months. This included:

1) *User Data:* Information on 10 user accounts (10 administrators)
2) *Student Data:* Records for 500 students, including personal details, enrollment status. (student ID, student name, status, grade, section, image, Address)
3) *Teacher Data:* Records for 50 teachers, including personal details, subjects taught, and class assignments. (Teacher ID, Teacher Name, status, image, Address)

B. Data Sources

The data was sourced from the school's existing administrative records, ensuring accuracy and relevance. Data privacy and security were maintained throughout the collection process, in compliance with ethical guidelines.

V. RESULTS

The evaluation of the School Management System (SMS) was conducted through a series of rigorous tests to ensure its functionality, reliability, and usability. The evaluation focused on the key functionalities of the system, including user authentication, dashboard displays, data entry, data update, data clear, and data deletion. The following sections outline the evaluation process and the results obtained from the tests conducted on the system.

A. *Test Data*

The test data used in the evaluation was collected from a mid-sized school over a period of three months.

B. *Test Scenarios*

1) *User Authentication:* Testing the login functionality to ensure that users can log in with valid credentials and are denied access with invalid credentials.
2) *Dashboard Functionality:* Testing the dashboard to ensure accurate display of Today enrolled student details and total student Teacher and Graduated student details.
3) *Data Entry:* Testing the addition of new student and teacher records to the database.
4) *Data Clear:* Clear the typing details test.
5) *Data Update:* Testing the updating of student and teacher records to ensure changes are correctly saved.
6) *Data Deletion:* Testing the deletion of teacher records to ensure records are correctly marked as deleted.

The results from the evaluation tests are summarized in the following sections.

C. *User Authentication*

1) *Success Rate:* 98% of logins were successful with valid credentials.
2) *Error Rate:* 2% of logins failed due to incorrect credentials or empty fields.
3) *Evaluation*: The login functionality is highly reliable, with only a minimal error rate.

D. *Dashboard Functionality*

1) *Accuracy:* 100% accurate display of enrolled student details.
2) *User Satisfaction:* Administrators reported high satisfaction with the ease of use and the information provided on the dashboard.
3) *Evaluation:* The dashboard provides a comprehensive and accurate overview of student details, meeting the needs of administrators effectively.

E. *Data Entry*

1) *Completion Rate:* 100% successful addition of student and teacher records.
2) *Error Rate:* Minimal errors, mostly due to user input mistakes.
3) *Evaluation:* The data entry functionality is robust, allowing for the seamless addition of new records with minimal errors.

F. *Data Update*

1) *Success Rate:* 99% successful updates of records.
2) *User Feedback:* Positive feedback from users on the ease of updating records.
3) *Evaluation:* The data update functionality is reliable and user-friendly, allowing users to make changes to records with confidence.

G. *Data Clear*

1) *Success Rate:* 99% successful clear of records.
2) *User Feedback:* Positive feedback from users on the ease of clear records field.

H. *Data Deletion*

1) *Success Rate:* 98% successful deletion of teacher records.
2) *Error Handling:* Effective error messages and handling for unsuccessful deletions.
3) *Evaluation:* The data deletion process is effective, with clear error handling to guide users in case of issues.

VI. CONCLUSION AND DISCUSSION

The School Management System developed in this project successfully streamlines administrative processes, enhancing efficiency and data management in schools. The evaluation results demonstrate the system's capability in handling user authentication, managing student and teacher data, and providing real-time updates and reports with high accuracy and reliability.

A. *Discussion*

1) *Efficiency and Accuracy:* The system significantly reduces the manual workload involved in administrative tasks, leading to greater efficiency and accuracy. The automated processes ensure that data entry and retrieval are performed with minimal errors, thus enhancing the overall quality of data management.

2) *User Feedback:* Initial user feedback from administrators and teachers has been positive, highlighting the system's user-

friendly interface and comprehensive functionalities. The training sessions and detailed documentation provided have further facilitated the smooth adoption of the system.

3) *Comparison with Existing Systems:* Compared to existing school management solutions, our SMS offers a more tailored approach with features specifically designed to address the unique needs of the schools involved in the study. Its cost-effectiveness and ease of integration with existing IT infrastructure make it a viable option for schools with limited resources.

4) *Limitations and Future Work:* While the system performs well in the pilot phase, there are areas for improvement. Future work will focus on enhancing the system's scalability to accommodate larger schools and integrating additional features such as advanced analytics and reporting tools. Furthermore, ongoing support and maintenance will be crucial in addressing any emerging issues and ensuring the system remains up-to-date with the latest technological advancements.

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